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Modularity and Evolvability of Simian Cranial Sexual Dimorphism

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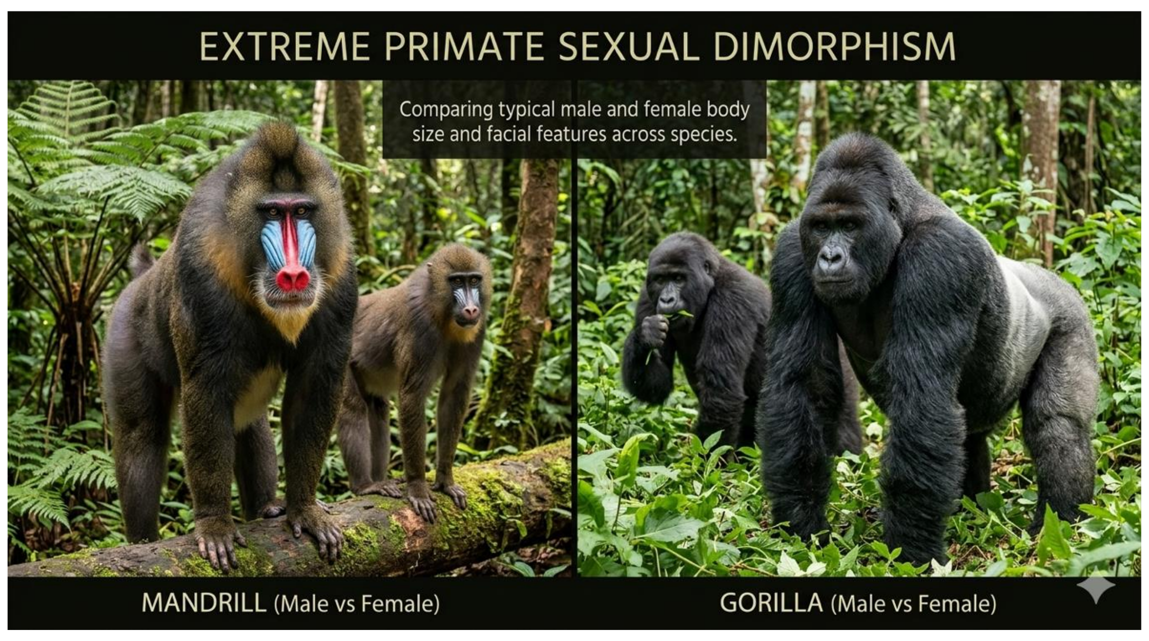
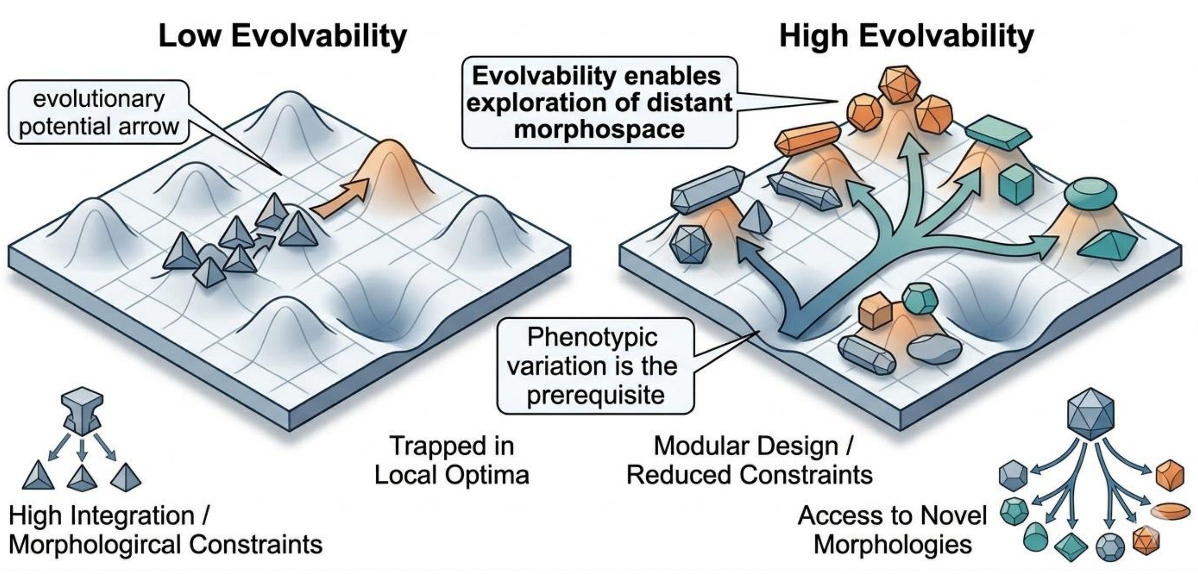
Sexual Dimorphism



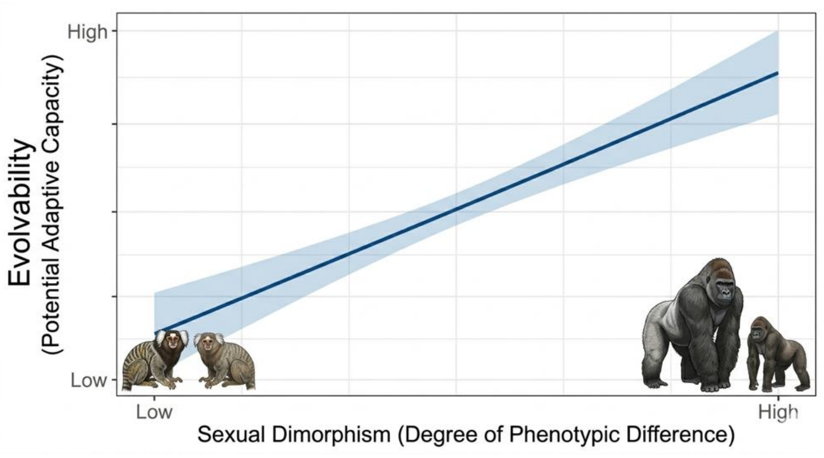
- Sexual dimorphism shows extensive interspecific variation
- The evolution of sexual dimorphism is likely multifactorial, but the role of evolvability remains unclear

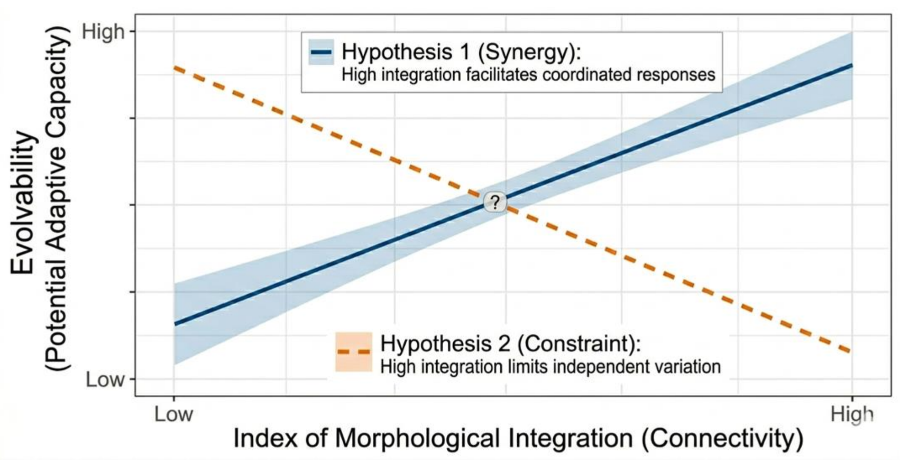
Evolvability and Modularity

- Low evolvability may constrain evolutionary change
- High evolvability may promote greater morphological diversification and faster evolutionary change



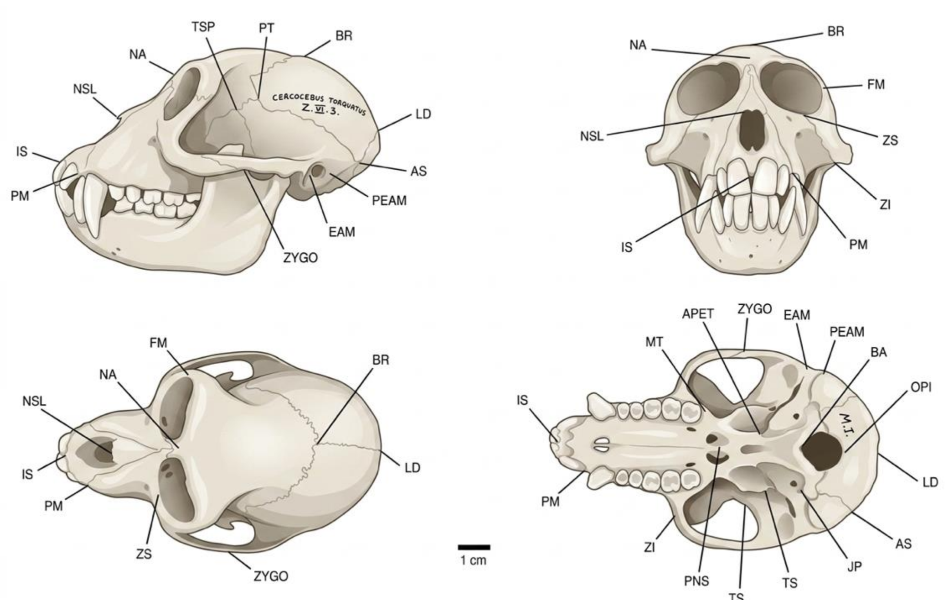
Hypothesis

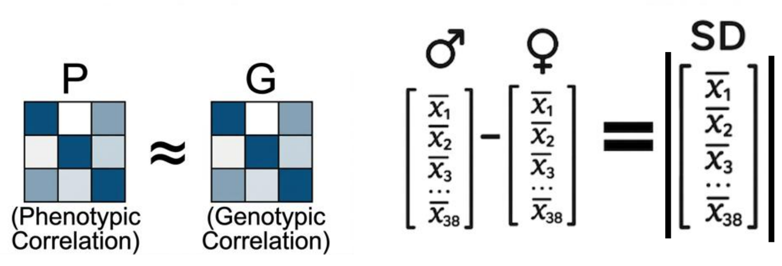




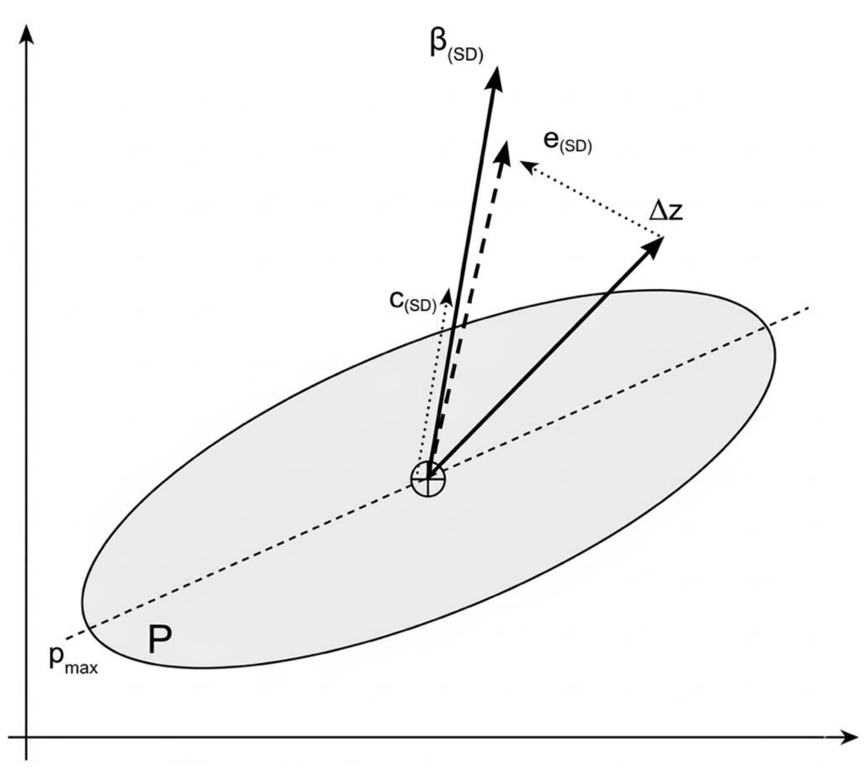
Data

- 50 species from two simian clades: Platyrrhini and Catarrhini
- 38 linear cranial measurements collected from simian skulls
- ~11,000 simian skulls measured.
- **P** matrices estimated for each species
- Sexual dimorphism was measured as the **norm** of the vector difference between males and females





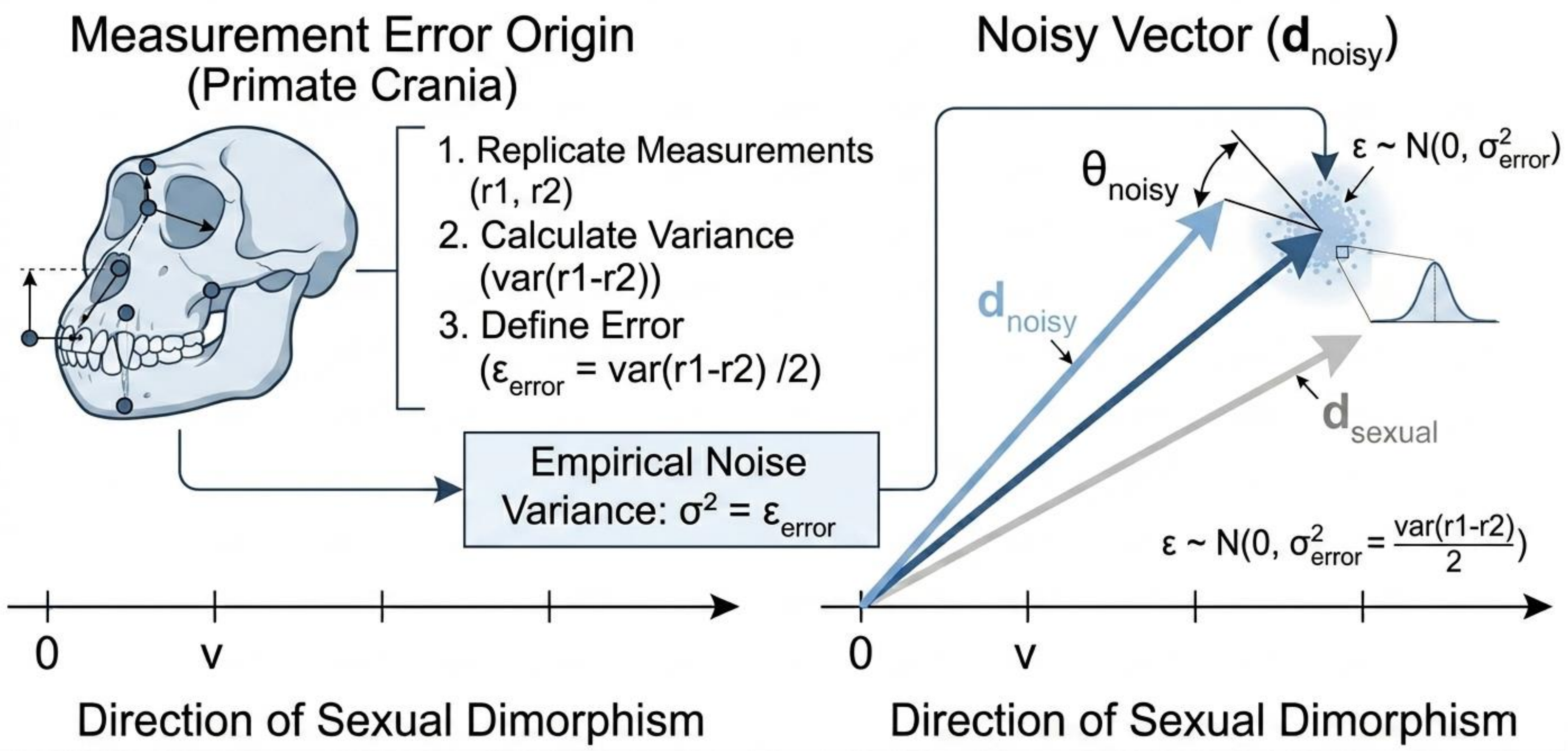
Analysis



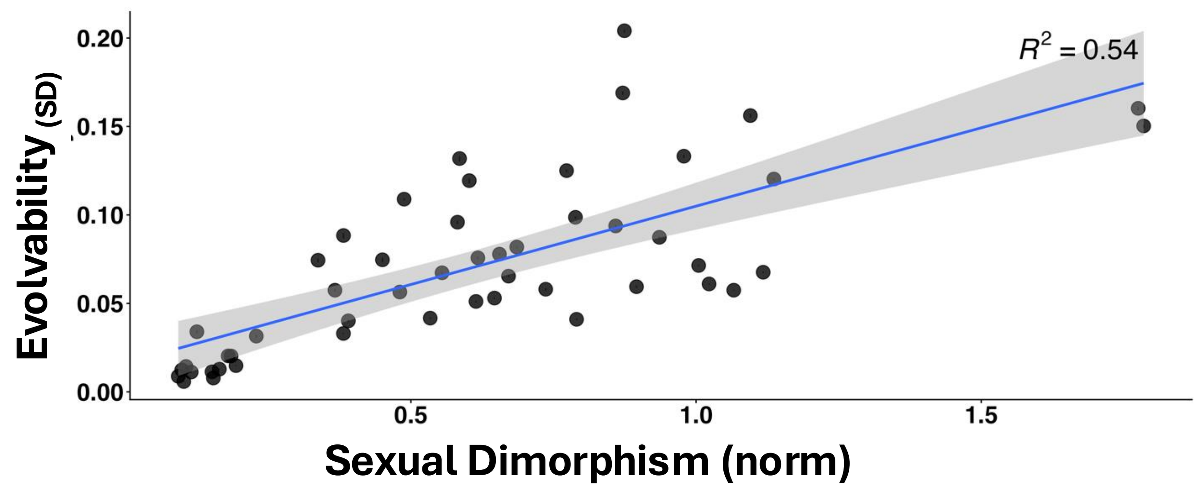
$$\text{evolvability}_{(\text{SD})} = \frac{(\text{SD} + \varepsilon)'\mathbf{P}\,\text{SD}}{|\text{SD} + \varepsilon|^2}$$

$$\text{integration} = 1 - \frac{1}{\text{SD}^{\text{T}}\mathbf{P}^{-1}\text{SD}}$$

$$\text{conditional evolvability}_{(\text{SD})} = \text{evolvability}_{(\text{SD})} \,.\, \text{integration}$$

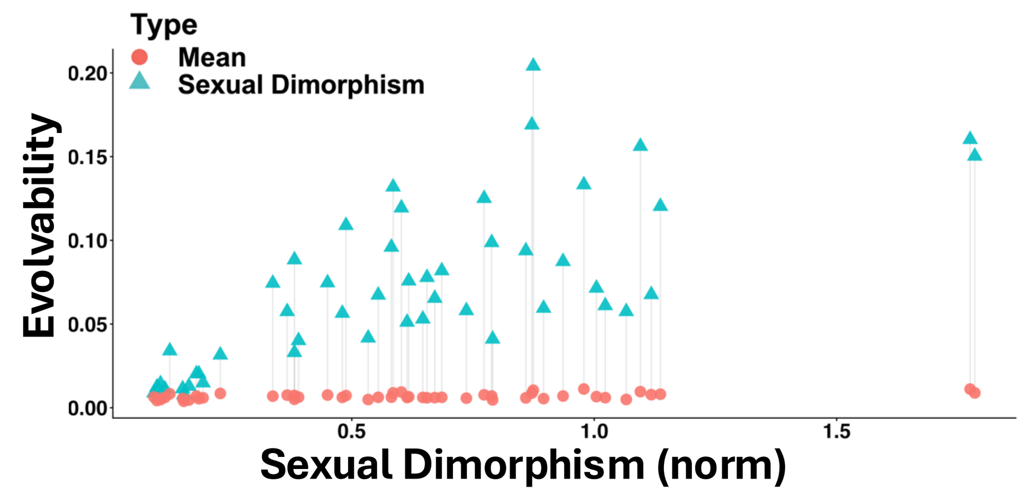


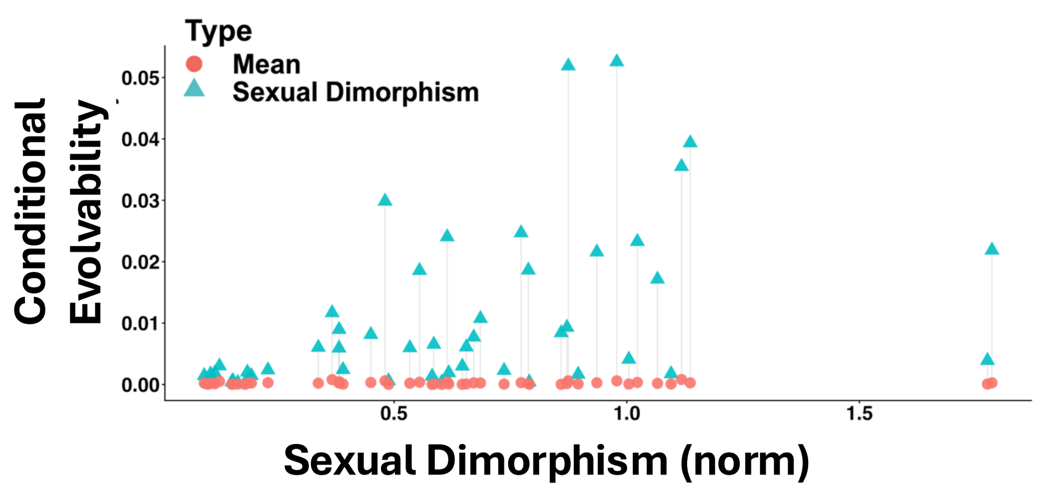
Results



- Greater sexual dimorphism is associated with higher evolvability

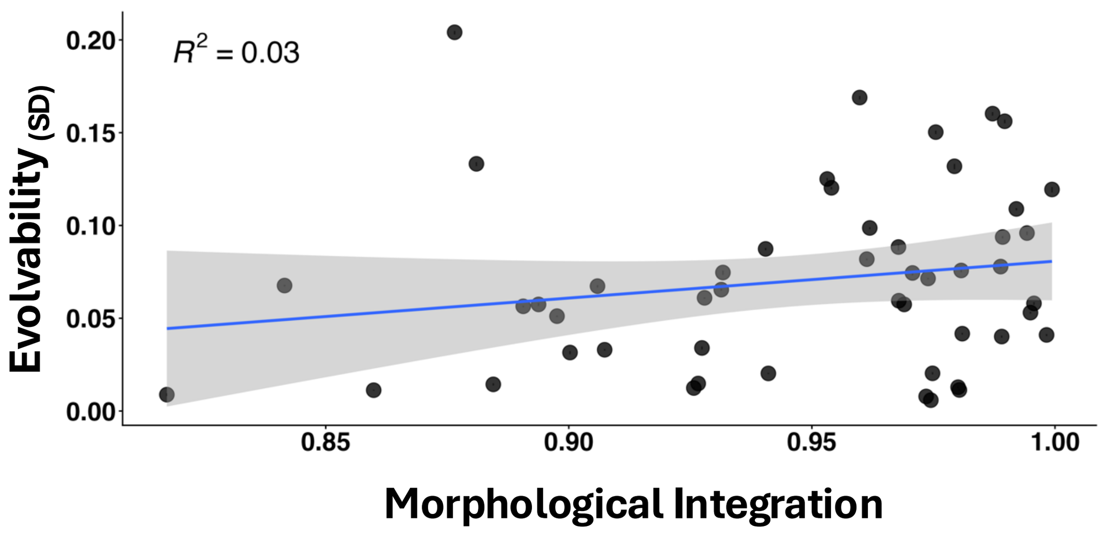
- Evolvability in the direction of sexual dimorphism is higher than the average evolvability

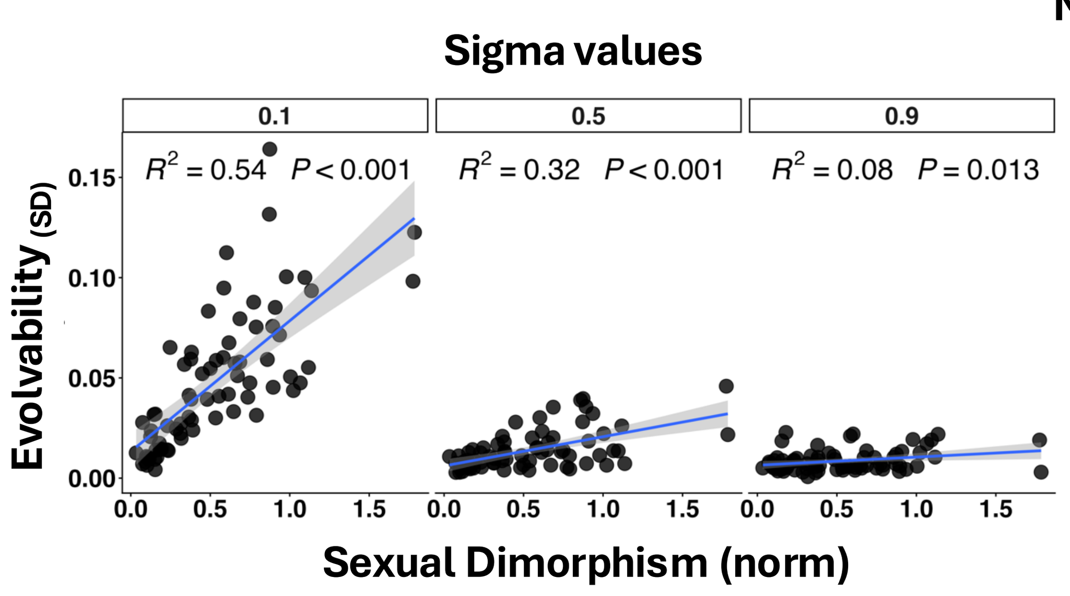




- Despite some variation, conditional evolvability in the direction of sexual dimorphism is higher than the average conditional evolvability

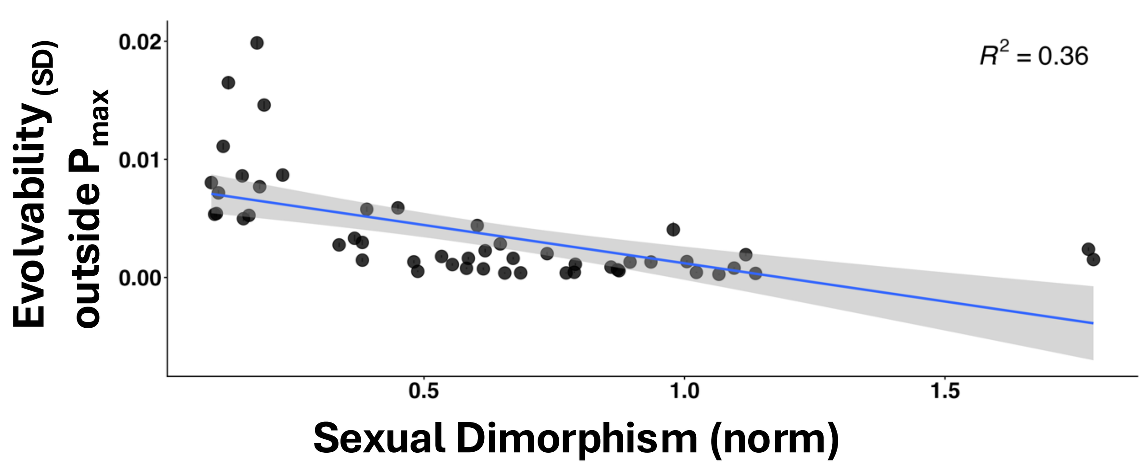
- No relationship was found between evolvability and morphological integration





- High measurement error weakens and may eliminate the relationship between sexual dimorphism and evolvability

- Excluding $\mathbf{P}_{max}$ changes the relationship, making it nonlinear and inverse



Conclusions and Next Steps

- Sexual dimorphism tends to evolve along highly evolvable directions
- Integration alone does not explain variation in evolvability among species
- Measurement error does not substantially alter the observed patterns
- Sexual dimorphism is frequently aligned with **Pmax**
- Identify additional predictors of evolvability in the direction of sexual dimorphism.
- Test whether these patterns are phylogenetically widespread
- Examine the consequences of removing **Pmax** on evolvability estimates.